

Professional Summary

Senior AI/ML Engineer and Data Scientist with 5+ years of hands-on experience in Python programming, Big Data technologies, cloud platforms (GCP, Azure), and enterprise-scale machine learning solutions. Demonstrated expertise in **Generative AI, LLMs, and agentic frameworks** (LangChain, RAG pipelines, vector databases), with deep proficiency in **MLOps, data engineering, and API development**. Proven track record of architecting and deploying production-grade AI/ML systems using **BigQuery, PySpark, Vertex AI, and serverless architectures**, driving measurable business impact through advanced analytics, time-series forecasting, and intelligent automation.

Career Objectives

- Design and implement enterprise-scale AI/ML solutions leveraging cloud platforms (GCP, Azure) and Big Data technologies to drive business transformation and innovation
- Build production-ready Generative AI applications using LLMs, agentic frameworks, and RAG pipelines to solve complex business challenges
- Deploy robust MLOps workflows integrating CI/CD, DevOps automation, and model monitoring to ensure reliable, scalable AI systems
- Develop advanced analytics and predictive models using Python, BigQuery, and distributed computing frameworks to extract actionable insights from large-scale datasets

Core Competencies

Python & Software Engineering: 5+ years of expert-level Python development with deep knowledge of OOP, data structures (queues, stacks, linked lists, trees, graphs), algorithms, and design patterns. Proficient in FastAPI, Flask, Django, and modern software engineering best practices including TDD, SOLID principles, and clean code architecture.

Big Data & Data Engineering: 4+ years of hands-on experience with BigQuery, PySpark, and distributed data processing. Expert in building scalable ETL/ELT pipelines, data warehousing, and real-time data streaming. Proficient with SQL, NoSQL (MongoDB, Firestore), pandas, numpy, and data modeling for large-scale analytics.

Cloud Platforms & AI/ML Services: 3+ years architecting solutions on GCP (BigQuery, Vertex AI, Cloud Run, Dataproc, Cloud Functions, Cloud Storage, IAP) and Azure (OpenAI, Cognitive Services). Expert in serverless architectures, Infrastructure as Code (Terraform), and cloud-native application development with Docker and Kubernetes.

Machine Learning & Deep Learning: Expert in ML model development using scikit-learn, XGBoost, TensorFlow, PyTorch, and advanced time-series forecasting (Autoformer, Informer, TFT). Proficient with Hugging Face Transformers, model optimization (Optuna), hyperparameter tuning, and feature engineering for production systems.

MLOps & Deployment: 4+ years implementing end-to-end MLOps pipelines with CI/CD automation (Airflow, Tekton, GitHub Actions), model versioning, A/B testing, monitoring, and deployment on cloud platforms (SageMaker, Vertex AI). Expert in containerization (Docker), orchestration, and scalable microservices architecture using Nginx & Unicorn.

Generative AI & LLMs: 3+ years of hands-on experience building production Generative AI applications with LLMs (OpenAI, Azure OpenAI, Gemini, Claude). Expert in agentic frameworks (LangChain, LangGraph), RAG pipelines, vector databases (Pinecone, Chroma), prompt engineering, and fine-tuning LLMs for domain-specific tasks.

API Development & DevOps: 5+ years designing and deploying RESTful APIs, API Gateways, and WebSocket

services at scale. Proficient in API security (OAuth2, JWT), rate limiting, load balancing, and DevOps automation with comprehensive monitoring, logging, and alerting (Prometheus, Grafana, CloudWatch).

Communication & Leadership: Excellent stakeholder management, technical writing, mentorship, and presentation skills. Proven ability to translate complex technical concepts for diverse audiences and lead cross-functional teams in Agile environments.

Professional Experience

Ford Motor Company

Detroit, MI, USA

Senior Data Scientist & AI/ML Engineer

Aug 2023 – Present

- Architected and deployed **enterprise-scale Python applications** using FastAPI and Flask, processing **100K+ data points/sec** with advanced OOP design patterns, data structures optimization, and RESTful API best practices for EV analytics microservices
- Designed and implemented **Big Data pipelines using PySpark and BigQuery**, ingesting and processing **10+ TB** of EV charging session data, enabling real-time analytics and predictive modeling for energy demand forecasting across Ford's Energy Service Simulator platform
- Led development of **Generative AI intelligent agent** using **LangChain and Ford LLM API** for automated code complexity analysis, providing time/space optimization recommendations and GCP cost estimations, reducing manual code review time by **60%**
- Deployed production **MLOps infrastructure on GCP** with Cloud Run, Vertex AI, and Dataproc, implementing **CI/CD pipelines via Tekton and Airflow**, achieving **99.9%** uptime and automated model retraining workflows with comprehensive monitoring
- Built **advanced ML models** (XGBoost, TFT, Autoformer, Informer) with Optuna hyperparameter optimization, improving parking probability predictions by **10%** and reducing training time by **40%** through distributed GPU computing on Vertex AI
- Engineered secure, scalable **data warehousing solutions in BigQuery** with partitioning, clustering, and query optimization strategies, reducing query costs by **35%** while enabling real-time dashboard analytics for business stakeholders
- Implemented **Infrastructure as Code using Terraform** for multi-environment (dev, QA, prod) GCP deployments, ensuring regulatory compliance, IAP authentication, and secure API Gateway configurations for enterprise applications

Key Projects:

- **EV Energy Optimization Platform:** Developed end-to-end Python application using FastAPI, BigQuery, and PySpark to predict energy bill savings for V2H/V2G participation; achieved **R-squared of 0.85** with XGBoost regression on 5+ million vehicle telematics data points
- **Generative AI Code Analyzer (Hackathon Winner):** Built intelligent agent using LangChain and LLMs to analyze GitHub repositories, estimate computational complexity, and recommend GCP Vertex AI configurations with cost optimization strategies
- **Real-Time EV Analytics Dashboard:** Created interactive Dash/Plotly dashboards with WebSocket integration, providing live insights into EV discharge opportunities and energy management for Ford's Command Center

CereVu Medical, Inc.

San Francisco Bay Area, CA

Software Engineer & AI/ML Researcher

Jun 2023 – Sep 2024

- Developed **production Python microservices** with Flask and FastAPI, implementing RESTful APIs and WebSocket endpoints for real-time biosignal streaming, supporting **10K+ concurrent connections** with sub-100ms latency
- Designed **MLOps CI/CD pipelines** using Tekton and AWS CodeDeploy, automating model deployment with Docker containerization, achieving **30%** faster release cycles and zero-downtime deployments with automated rollback mechanisms
- Built scalable **data engineering infrastructure** with MongoDB and AWS (S3, EC2), implementing ETL pipelines for processing biosignal data streams, enabling real-time analytics and machine learning model training
- Deployed **deep learning models** (LSTM, CNN) using Nginx & Unicorn on AWS, improving patient monitoring accuracy by **15%** through advanced signal processing and feature extraction algorithms
- Architected full-stack iOS application in SwiftUI with Firebase/Firestore integration, implementing secure authentication (OAuth2/JWT), real-time data synchronization, and cloud-based health data management

Applied Cognitive Ergonomics Lab, Texas A&M University

College Station, TX

Research Assistant & ML Engineer

Sep 2021 – May 2024

- Designed and implemented **end-to-end Python ML pipelines** for wearable-based stress detection using TensorFlow and PyTorch, achieving **76%** accuracy improvement through advanced feature engineering, sensor fusion, and deep learning architectures (LSTM, CNN)
- Built production iOS application (*GlucoseCoach*) in SwiftUI with Python Flask backend, implementing real-time ML inference, data synchronization, and personalized health monitoring, improving patient adherence metrics
- Developed **computer vision ML models** in Python for COVID-19 compliance monitoring, achieving **98%** accuracy in mask detection and social distancing using OpenCV, TensorFlow, and object detection algorithms
- Applied explainable AI techniques (SHAP, LIME) to ML models, enhancing clinical interpretability and stakeholder trust in healthcare decision-making systems

Department of Psychological & Brain Sciences, Texas A&M University

College Station, TX

Research Assistant & Data Scientist

Sep 2018 – Aug 2021

- Developed *OpenSync*, a **Python-based data integration platform** with WebSocket APIs for real-time synchronization of EEG/ERP and biosensor data streams, processing multi-modal behavioral data for neuroscience research
- Implemented **advanced signal processing algorithms** in Python (scipy, numpy) for EEG data preprocessing, filtering, and feature extraction, improving data quality and enabling sophisticated cognitive analysis
- Built **deep learning models** (Conv-LSTM) in TensorFlow for gait analysis, achieving **93%** accuracy in anomaly detection and fall risk prediction through time-series analysis of sensor data
- Engineered data pipelines integrating multiple biosensors, accelerating research workflows and enabling large-scale cognitive workload assessments across laboratory studies

Education

Texas A&M University

College Station, TX

Ph.D. in Industrial Engineering (Data Science)

Aug 2021 – Dec 2023

Dissertation: "Advancing Stress Detection: Real-Time, Naturalistic, and Personalized AI Approaches for Stress and Mental Disorder Detection"

GPA: 4.0/4.0 *Summa Cum Laude*

Texas A&M University

College Station, TX

M.S. in Computer Science

Aug 2019 – May 2021

Relevant Coursework: Machine Learning, Deep Learning, NLP, Algorithms, Pattern Recognition, Computer Vision, Database Systems, Reinforcement Learning, Data Mining

GPA: 4.0/4.0 *Summa Cum Laude*

Technical Skills

Programming Languages: Python (Expert), Java (Spring Boot), JavaScript/TypeScript (React, Next.js, Node.js), Swift (SwiftUI), SQL, Bash

Python Frameworks & Libraries: FastAPI, Flask, Django, pandas, numpy, scipy, scikit-learn, XGBoost, TensorFlow, PyTorch, Hugging Face Transformers, LangChain, LangGraph, PySpark, Optuna, Streamlit, Dash, Plotly

Big Data Technologies: BigQuery, PySpark, Apache Hadoop, Apache Airflow, Dataproc, data warehousing, ETL/ELT pipelines, distributed computing, real-time streaming, data partitioning & clustering

Cloud Platforms: **GCP:** BigQuery, Vertex AI, Cloud Run, Cloud Functions, Dataproc, Cloud Storage, IAP, Pub/Sub, Cloud Logging, Cloud Monitoring

Azure: OpenAI Service, Cognitive Services, Azure Functions

AWS: S3, EC2, Lambda, SageMaker, CodeDeploy, CloudWatch, Auto Scaling

Generative AI & LLMs: OpenAI (GPT-4, GPT-3.5), Azure OpenAI, Google Gemini, Anthropic Claude, LangChain, LangGraph, RAG pipelines, vector databases (Pinecone, Chroma), prompt engineering, fine-tuning, agentic frameworks

MLOps & DevOps: Docker, Kubernetes, Terraform (IaC), Tekton, Apache Airflow, GitHub Actions, CI/CD pipelines, model versioning, A/B testing, monitoring (Prometheus, Grafana), Nginx, Gunicorn, load balancing

Databases: BigQuery, MongoDB, Firestore, PostgreSQL, MySQL, SQLAlchemy, NoSQL data modeling, query optimization, database design

API & Web Development: RESTful APIs, API Gateway, WebSocket, OAuth2, JWT, FastAPI, Flask, Express.js,

React, Next.js, Material-UI, Redux, GraphQL, OpenAPI/Swagger

ML/DL Frameworks: Scikit-learn, XGBoost, TensorFlow, PyTorch, Keras, Hugging Face Transformers, time-series forecasting (Autoformer, Informer, TFT), LSTM, CNN, computer vision, NLP

Data Science & Analytics: Statistical analysis, A/B testing, feature engineering, model evaluation, hyperparameter tuning (Optuna), data visualization (Matplotlib, Seaborn, Plotly), Jupyter Notebooks

Software Engineering: OOP, design patterns, data structures & algorithms, SOLID principles, TDD, Git, Agile/Scrum, code review, system design, microservices architecture

Certifications

- **Advanced MLOps Specialization** (Coursera) – End-to-end MLOps best practices with real-world applications
- **NLP Specialization** (Coursera) – Advanced NLP techniques & large-scale language modeling
- **Deep Learning Specialization** (DeepLearning.AI) – Fundamentals & advanced neural network architectures
- **Bayesian Statistics** (Coursera) – Probabilistic modeling & inference for data-driven decisions
- **Generative AI with LLMs** (AWS) – Practical, hands-on generative AI methodologies

Selected Publications

2024: Razavi, M., et al. *Integrated Eye-Tracking and EEG Data Collection and Synchronization for VR-Based Spatial Ability Assessments*. *IHSI*

2023: Razavi, M., et al. *Machine Learning, Deep Learning, and Data Preprocessing for Stress Detection: A Scoping Review*. *JMIR*

2023: Janfaza, V., Weston, K., Razavi, M., et al. *MERCURY: Accelerating DNN Training via Input Similarity*. *IEEE HPCA*

2023: Kim, N., Grégoire, L., Razavi, M., et al. *Virtual Accident Curb Risk Habituation in Workers*. *iScience*

2022: Razavi, M., et al. *Automatic Monitoring of Physical Distancing & Mask Wearing*. *SN Computer Science*

2021: Razavi, M., et al. *OpenSync: Synchronizing Multiple Measures in Neuroscience*. *Journal of Neuroscience Methods*